ZIRAHR: Time & Attendance (T&A)

(with Geofencing & Overtime)

Objective

Build a **smart, mobile-first attendance system** that:

- Allows employees to **clock in/out via self-service portal**
- Uses **geofencing for location validation**
- Stores attendance in a **centralized daily record**
- Provides **real-time visibility to supervisors**
- Supports **automated + approval-based overtime**
- Fully configurable via **UI settings (no hardcoding)**

END-TO-END FLOW (CRITICAL)

1. Employee logs in (Self-Service Portal)
2. Clicks **Clock In / Clock Out**
3. System:
    - Captures **timestamp**
    - Captures **GPS location**
    - Validates against **geofence (if enabled)**
4. Record stored in **Attendance Table (centralized)**
5. Data:
    - Visible to employee
    - Visible to assigned supervisor
    - Synced to correct **branch/workspace**
6. System computes:
    - Working hours
    - Late/early flags
    - Overtime (if applicable)
7. Overtime:
    - Auto-approved OR
    - Routed for approval (1–2 approvers max)

EPIC 1: Employee Clock-In / Clock-Out (Self-Service)

Epic	Feature	User Story	Acceptance Criteria
Attendance	Clock In	As employee, I clock in	Timestamp recorded
Attendance	Clock Out	As employee, I clock out	Timestamp recorded
Attendance	Status Display	As employee, I see my status	"Checked In / Checked Out" visible
Attendance	Daily Record	As employee, I see my logs	Day summary visible

Mobile UX Requirements

- One-click **Clock In / Out**
- Clear status:
    - "You are checked in since 08:03 AM"
- Prevent duplicate clock-ins
- Offline handling (queue + sync later)

EPIC 2: Geofencing & Location Validation

Epic	Feature	User Story	Acceptance Criteria
Geo	Location Capture	As system, I capture GPS	Lat/Long stored
Geo	Geofence Validation	As system, I validate location	Check within allowed radius
Geo	Restriction	As HR, I restrict access	Block if outside zone (optional)
Geo	Multi-Location	As system, I support branches	Different geofences per branch

Geofence Logic

- Each **branch/workspace** has:
    - Latitude
    - Longitude
    - Radius (e.g., 100m)

If outside:

- Either:
    - Block clock-in
    - OR allow but flag as “Outside Location”

EPIC 3: Centralized Attendance Storage

Epic	Feature	User Story	Acceptance Criteria
Storage	Daily Records	As system, I store logs	One record per user/day
Storage	Time Logs	As system, I store times	Check-in/out timestamps saved
Storage	Location Logs	As system, I store GPS	Location tied to record
Storage	Status	As system, I compute status	Present/Absent/Late

Data Model (Simplified)

attendance_records

- user_id
- date
- check_in_time
- check_out_time
- total_hours
- overtime_hours
- status
- branch_id
- location_in (lat/long)
- location_out (lat/long)

EPIC 4: Supervisor Visibility & Control

Epic	Feature	User Story	Acceptance Criteria
Visibility	Team Attendance	As supervisor, I see my team	Filter by assigned employees

Visibility	Daily Dashboard	As supervisor, I see today	List of employees + status
Visibility	Drill Down	As supervisor, I view details	Click → full record

Supervisor View

- Employee Name
- Check-In Time
- Check-Out Time
- Status (Late, Absent, Present)
- Overtime Hours

Only shows employees **assigned to that supervisor**

EPIC 5: Branch / Workspace Sync

Epic	Feature	User Story	Acceptance Criteria
Sync	Branch Assignment	As system, I assign branch	Each employee linked to branch
Sync	Data Segmentation	As system, I segment data	Attendance filtered by branch
Sync	Multi-Workspace	As system, I isolate data	Workspace-level separation

Logic

- Employee → belongs to → Branch
- Attendance → tagged with → Branch

Ensures:

- Nairobi data ≠ Mombasa data
- Clean reporting

EPIC 6: Working Hours & Organization Settings (CONFIGURABLE)

Epic	Feature	User Story	Acceptance Criteria
Settings	Working Days	As HR, I define workdays	e.g., Mon–Fri
Settings	Working Hours	As HR, I define hours	Start & End time
Settings	Non-Working Days	As HR, I define holidays	Days excluded
Settings	Late Rules	As HR, I define lateness	Grace period configurable

Example Config

- Start Time: 08:00 AM
- End Time: 05:00 PM
- Grace Period: 15 mins
- Workdays: Mon–Fri

EPIC 7: Overtime Computation (Kenya-Compliant)

Epic	Feature	User Story	Acceptance Criteria
Overtime	Auto Calculation	As system, I compute OT	Hours beyond work time
Overtime	OT Types	As HR, I define OT types	Ordinary vs Special
Overtime	Multipliers	As system, I apply rates	1.5x / 2x applied
Overtime	Threshold	As system, I trigger OT	After working hours

🇰🇪 Overtime Types

Type	Rule
Ordinary OT	1.5x (after working hours)
Special OT	2x (weekends/public holidays)

EPIC 8: Overtime Approval Workflow

Epic	Feature	User Story	Acceptance Criteria
Approval	Toggle Approval	As HR, I enable approval	ON/OFF toggle
Approval	Assign Approvers	As HR, I assign approvers	Max 2 approvers
Approval	Approval Flow	As approver, I review OT	Approve/Reject
Approval	Status Tracking	As system, I track OT	Pending/Approved/Rejected

Approval Logic

- If OFF → Auto-approved
- If ON →
    - Level 1 Approver
    - Level 2 (optional)

EPIC 9: Reporting & Export

Epic	Feature	User Story	Acceptance Criteria
Reports	Daily Report	As HR, I see attendance	Per day view
Reports	Monthly Report	As HR, I see summary	Per employee totals
Reports	Export	As HR, I export data	CSV/Excel download

EPIC 10: Notifications & Alerts

Epic	Feature	User Story	Acceptance Criteria
Alerts	Missed Clock-In	As system, I alert user	Notification sent
Alerts	Late Arrival	As system, I flag lateness	Visible in dashboard
Alerts	OT Approval	As approver, I get alert	Notification triggered

UI CONFIGURATION PANEL (VERY IMPORTANT)

Everything must be configurable:

Attendance Settings UI

- Working Days
- Start Time / End Time
- Grace Period
- Geofencing:
  - Enable/Disable
  - Radius
- Overtime:
  - Enable/Disable
  - Approval Toggle
  - Approvers (max 2)
  - OT Types (1.5x / 2x editable)

NO hardcoded logic.

Developer Advisory (CRITICAL)

Performance

- Use **batch processing for daily summaries**
- Avoid recalculating on every request

Accuracy

- Store **raw timestamps** (don't overwrite)
- Compute derived values separately

Geolocation

- Use **browser GPS API**
- Handle:
  - Permission denied
  - Low accuracy

Mobile

- Optimize for:
  - Slow networks
  - Offline fallback

Security

- Prevent:
  - Fake timestamps
  - Location spoofing (basic checks)

SYSTEM INTEGRATION

- **Payroll Module (Future):**
  - Uses attendance + OT for salary
- **HR Module:**
  - Links to employee profile
- **Branch System:**
  - Filters data by workspace

Final Positioning

This module becomes:

The source of truth for workforce presence
The **input layer for payroll & compliance**

Final Note to Developer

- Clock-in must be **1 click**
- Response must be **instant (<1 sec)**
- UI must be **clean and not confusing**
- System must **never lose data**

If employees struggle to clock in, the entire system fails.